
Verification-Gated Autonomous Optimization of Systems Code

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Abstract

1 We study whether an autonomous, LLM-driven optimization loop can discover
2 algorithmic improvements to a production consensus library while preserving
3 every safety property of the protocol. Our target is `etcd-io/raft`, the consensus
4 library beneath Kubernetes and CockroachDB. We build a layered verification
5 harness around it: an existing TLA^+ specification for protocol-level invariants,
6 a deterministic fault-injection simulator that drives the library through partition,
7 drop, reorder, and duplication scenarios, and a linearizability check on the resulting
8 client histories. The harness then gates an evolutionary search driven by a frontier
9 coding model: every candidate edit must clear compilation, the fault sweep, and
10 the linearizability check before its benchmark score is even reported. Our central
11 claim is methodological: where safety is mechanically checkable, the harness
12 (not the model) is the load-bearing artefact, and a sufficiently sharp harness lets
13 a general-purpose coding agent search inside a mature systems codebase safely.
14 Harness-acceptance is necessary but not sufficient for upstream merge; we make
15 no production-readiness claim.

16 1 Introduction

17 Raft [24] is the consensus algorithm at the core of most contemporary distributed systems: `etcd` [10],
18 Kubernetes [27], CockroachDB [8], Docker Swarm [9], TiKV [29]. Implementations of Raft are
19 required to preserve five safety properties (election safety, log matching, leader completeness, state-
20 machine safety, and log durability under partial failure) under every interleaving of message loss,
21 reorder, duplication, and crash-restart. The cost of an undetected regression is, in practice, total: a
22 single safety-violating commit can corrupt the entire replicated state machine across operators.

23 Recent work on AI-driven research for systems [5, 6, 16] has shown that LLM-driven evolutionary
24 search can match or exceed human-tuned performance on isolated kernels, schedulers, and codec
25 libraries when correctness is verifiable. The natural question is whether the same loop can be turned
26 on a production-grade consensus library, where correctness is a global protocol invariant rather than a
27 per-operation predicate.

28 Our hypothesis is that the bottleneck is verification bandwidth, not generation quality. Concretely:
29 if every candidate variant produced by the LLM is forced through a mechanical pipeline (compile,
30 seeded deterministic fault-injection sweep, linearizability check, benchmark) before any performance
31 score is reported, the generator can be allowed to explore aggressively without endangering the
32 codebase. The harness, not the model, becomes the artefact whose quality determines what kinds of
33 improvements are reachable.

34 This paper instantiates that pattern over `etcd-io/raft` [11] at a pinned commit. Our contributions
35 are:

1. A minimal-upstream-patch deterministic simulation testing (DST) [32, 33] harness over the library’s existing `RawNode/Storage` interfaces, with seeded fault injection (partition, heal, drop, reorder, duplicate) and a three-layer oracle (TLA⁺ trace validation, Raft-protocol invariants and Porcupine linearizability on client histories). The harness achieves bit-for-bit reproducibility per seed after a four-line patch to the upstream election-timeout random number generator (RNG).
 2. Calibration of harness noise floors: we report 0 invariant violations across 200 distinct fault-injected seeds on the unmodified upstream, a 17% Porcupine false-positive rate attributable to tick-level read-resolution timing, and a bimodal benchmark distribution.
 3. An evolutionary-search wrapper that uses the harness as the rejection oracle. A frontier coding model proposes bounded edits to the leader-side flow-control function `maybeSendAppend`; candidates that fail to compile, fail any upstream test, fail any fault-injection scenario, or produce non-linearizable client histories beyond the noise floor are discarded before benchmarking.
 4. An empirical study reporting accepted variants, their throughput characteristics against the pinned baseline, and an ablation indicating which harness layer first catches which class of incorrect candidate.
- Non-goals. We do not retrain or fine-tune the underlying model. We do not propose accepted variants as production-ready: harness acceptance is necessary but not sufficient for upstream merge, and human review and shadow deployment remain in scope.

2 Background

2.1 Raft

Raft [24] is a leader-based consensus protocol over a cluster of replicas. A unique leader per term linearizes all proposals via an append-only replicated log; a majority of replicas must persist an entry before it is committed; followers apply committed entries to an arbitrary state machine in log order. The protocol’s safety rests on five invariants: *election safety* (at most one leader per term); *leader append-only*; *log matching* (identical prefixes across replicas); *leader completeness* (any committed entry appears in every subsequent leader log); and *state-machine safety* (no two replicas apply divergent commands at the same index). We treat all five as required invariants in our harness.

2.2 The `etcd-io/raft` Library

`etcd-io/raft` is a Go library implementing Raft as a pure state machine: it owns no network sockets, no disk, and no goroutine loops. The public `Node/RawNode` surface exposes `Tick()`, `Step(m)`, `Propose`, `ReadIndex`, `Campaign`, `Ready()`, and `Advance`; a pluggable `Storage` interface provides log persistence. This design makes the library unusually amenable to deterministic simulation testing: ticks are logical, message delivery and disk are caller-controlled, and the state machine’s output is fully determined by its input sequence, given that the few sources of non-determinism inside the library are themselves controllable.

The repository ships three artefacts we reuse. (1) A `rafttest` package providing a command-driven interaction environment for data-driven tests; we adopt its `InteractionEnv` only as a reference, building our own scheduler atop `RawNode` for tighter control. (2) A `state_trace.go` file under a `with_tla` build tag emitting `TracingEvent` records at every state transition. (3) A `tla/` directory containing `MCetcdraft.tla` (the model-check spec) and `Traceetcdraft.tla` (a trace-validation spec consuming the NDJSON output of `state_trace.go`).

2.3 Harness-First and Verification-Gated Search

The harness-first thesis [16] inverts the model-vs-review tradeoff: instead of asking whether agent-written code is correct, a layered set of automated oracles asks whether the candidate falsifies any property. Prior instantiations target a Redis-compatible store [23] and a Kafka-compatible engine [16]. Verification-gated autonomous optimization [5, 6] extends the pattern with an LLM candidate generator and an evolutionary database [3, 22, 25, 26], gated on a formal or empirical

85 oracle; reported deployments have driven large speedups on time-series codecs behind Verus [19]
 86 safety proofs. We instantiate the pattern over a Go consensus library: the harness combines TLA⁺
 87 trace validation, deterministic simulation, and linearizability checking, and the search target is a
 88 single function inside a ~85 KB host file.

89 3 Verification Harness

90 3.1 Design Principles

91 We adopt three principles. *Independence*: each layer should fail for a different reason: a wrong
 92 protocol spec passes deterministic simulation, an incomplete fault catalogue passes the spec, a
 93 permissive state-machine model accepts non-linearizable histories that the spec rejects. *Reuse-first*:
 94 where upstream already ships verification infrastructure (the `state_trace.go` hooks and the in-tree
 95 TLA⁺ files), we wrap rather than reimplement. *Cheap iteration*: every layer must terminate in
 96 seconds per scenario, since the optimization loop in §4 will invoke the full pyramid hundreds of times
 97 per run.

98 We achieve bit-for-bit reproducibility per seed via a single four-line patch to the upstream library:
 99 a new optional `Config.Rand` field that, when set, replaces a `crypto/rand`-backed source for the
 100 randomized election timeout (the only non-seedable randomness in the library, identified by a single
 101 `globalRand.Intn` call in `raft.go`). When the field is nil, the existing behaviour is preserved; the
 102 upstream test suite continues to pass without modification.

103 3.2 Layer 1: Protocol Specification

104 For the protocol-level oracle, we reuse the in-tree `MCetcdraft.tla` spec and the
 105 `Traceetcdraft.tla` trace-validation spec. The harness is built with `-tags=with_tla` so that
 106 `state_trace.go` emits a `TracingEvent` stream on every state transition (election, becoming
 107 leader, sending append entries, etc.). A run’s NDJSON trace is then validated against the TLC model
 108 checker via the upstream `tla/validate.sh` script; any state or transition not admitted by the spec
 109 is a failure.

110 3.3 Layer 2: Deterministic Simulation Testing

111 The DST layer is a from-scratch single-goroutine scheduler built atop `RawNode`; it advances the
 112 cluster one logical tick at a time in four phases. **Clocks**: every healthy node is ticked, possibly firing
 113 an election or a heartbeat. **Progress**: each node drains its `Ready()`; the harness persists the new
 114 entries to a per-node `MemoryStorage`, applies committed entries to a single-register state machine,
 115 and enqueues outbound messages onto the network in a canonical order. A deterministic sort is
 116 required because upstream builds the outbox by iterating a Go map. **Delivery**: messages whose
 117 scheduled-delivery tick has arrived are `Step()`-ed into their destination, possibly being dropped,
 118 reordered, or duplicated by the network layer. **Reads**: pending client reads complete once the
 119 linearizable safe index has caught up. All randomness (ours and upstream’s election timeout) flows
 120 from a single seeded `*rand.Rand`, exposed via the four-line patch of §3; full pseudocode and the
 121 bindings to upstream’s `RawNode/Storage` interface are in Appendix A.1.

122 The network supports per-link drop probability, partition/heal, per-message scheduled-delivery delay
 123 (yielding reorder when delays exceed one tick), and per-send duplication. The default scenario applies
 124 $p_{\text{drop}} = 0.02$ on every directed link, $p_{\text{partition}} = 0.01$ per tick, $p_{\text{heal}} = 0.05$, $\text{maxDelay} = 3$ ticks, and
 125 $p_{\text{dup}} = 0.02$.

126 The invariant oracle checks two properties on every tick: *log-prefix agreement* (for any two nodes
 127 i, j and any index k present in both node’s committed list, the entries must be identical) and *leader*
 128 *uniqueness per term* (no two nodes are in `StateLeader` at the same term).

129 3.4 Layer 3: Linearizability

130 We adopt the classical linearizability correctness condition [14] on the sequence of completed client
 131 operations. Three logical clients, each preferring a distinct cluster node, issue writes via `Propose` and
 132 linearizable reads via `ReadIndex` into a single-register state machine. Each client keeps at most one

operation in flight; writes encode a 17-byte payload so committed entries can be matched back to the issuing client. Reads are finalised once the preferred node’s `appliedIndex` crosses the safe index returned in `Ready.ReadStates`; the read returns the register snapshot captured at the safe index (rather than the current register), which is necessary because multiple entries may apply in a single Ready cycle. The recorded (invoke, return) history is then checked against a register specification using Porcupine [2], a Go re-implementation of the checker pattern introduced by Knossos [18] for Jepsen [17].

3.5 Fault Catalogue and Coverage

The catalogue is intentionally small but distinguishing: partition, heal, per-link drop, per-message delay/reorder, and per-send duplication. Crash-restart is excluded by construction because our `MemoryStorage` does not persist conf state without snapshot support; we treat this as a deferred extension and discuss it in §7.

3.6 Layer Costs

On a 16-vCPU x86_64 cloud host (32 GB RAM, Ubuntu 24.04), single-process, the per-candidate cost of the harness pyramid (compile, 8 fault-injected DST seeds at 1500 ticks each, Porcupine check, and a five-run benchmark) is approximately 15–20 seconds wall-clock; the LLM call dominates total per-iteration time. The DST layer alone runs at ~12 k ticks/second for a three-node cluster; a 200-seed sweep at 2000 ticks completes in under four minutes.

4 Autonomous Optimization Loop

4.1 Search Space

We restrict the optimizer to one function in `raft.go`, the leader’s per-follower send decision `maybeSendAppend(to, sendIfEmpty)`. It runs once per follower per tick and decides whether to emit an `AppendEntries` message, governing flow control, batching, pipelining, and snapshot fallback, the hottest function on the leader’s outbound path (~45 lines). We chose it because every throughput lever is local to the function, yet a wrong edit can break log consistency, making it a sharp test of the harness rather than the generator. The levers in scope (a throttle check, the replication-state predicate, the owed-entries log slice, the post-send progress update, and the snapshot fallback) map to their upstream identifiers in Table 2. The mutable region is delimited by `EVOLVE-BLOCK` markers; diffs touching the file outside it are rejected before compilation.

The generator is instructed to preserve five behavioural contracts, stated by purpose: the signature; the throttle short-circuit (never send to a paused or window-full follower); snapshot fallback when the required log prefix has been compacted; the post-send progress update (else later ticks break log matching); and no new package-level state, goroutines, locks, or I/O. The middle three are exactly what the harness re-checks mechanically: the prompt states them so the generator need not rediscover them, but the harness, not the prompt, enforces them. The verbatim prompt is in Appendix A.6.

4.2 Candidate Generator

A frontier coding agent receives the entire host file plus the system prompt above and is asked to emit a `SEARCH/REPLACE` diff scoped to the marked region. The generator is invoked through an OpenAI-compatible chat endpoint and is, by configuration, never informed of the benchmark scores produced by previous candidates; only that they were “accepted” or “rejected” by the harness, with diagnostics from the failing layer if any. This design keeps the generator from over-fitting to bench noise.

4.3 Rejection Oracle

The rejection oracle is the harness pyramid of §3, applied in order of increasing cost: (a) upstream package compilation; (b) the DST sweep ($N = 8$ seeds, 1500 ticks each, full fault catalogue); (c) the linearizability check on the recorded client histories; only then (d) the benchmark. Any layer’s failure terminates evaluation immediately and is recorded as the candidate’s rejection reason. The

harness layers contribute failure modes independently: a candidate that compiles correctly but violates log-matching is caught at (b), while a candidate that passes (b) but reorders `Ready.Messages` in a way that breaks linearizability is caught at (c). The unmodified upstream test suite is *not* run per candidate (it adds ~ 5 seconds for negligible additional coverage given (b)) but is run on accepted candidates as part of the post-acceptance re-validation described in §6.3.

4.4 Candidate Database and Selection

We use a published open-source evolutionary search framework [25] that maintains a SQLite-backed candidate database with MAP-Elites-style island distribution. Parent sampling is power-law over accepted candidates; mutation is the LLM’s SEARCH/REPLACE diff. The framework retains rejected candidates with their rejection-reason metadata, which is consumed by the per-layer ablation in §6.

4.5 Scoring

A candidate that clears the oracle is scored as

$$s = \frac{n_{\text{op}}^{\text{base}}}{\min_{i=1\dots K} n_{\text{op}}^{\text{cand},i}}$$

where $n_{\text{op}}^{\text{cand},i}$ is the per-run ns/op of the upstream `BenchmarkProposal3Nodes` on the candidate, $K = 5$ runs are taken to minimise jitter sensitivity, and $n_{\text{op}}^{\text{base}}$ is a fixed calibration constant chosen so the unmodified baseline scores near 1.0. Scores below the calibrated noise floor are not interpretable as improvements; see §6 for the floor measurement.

5 Experimental Setup

Target. `etcd-io/raft` pinned at commit 67d129e8 (head of `main` on 2026-05-26). The single patch to upstream is the seedable election-timeout RNG of §3; all other modifications live in a separate Go module and are applied to a working copy of `raft.go` on each evaluation. The unmodified upstream test suite is run on accepted candidates as part of post-acceptance re-validation.

Software and hardware. Go 1.26.3; OpenEvolve [25] 0.2.27; Porcupine [2] v1.1.0; OpenJDK 17 with `tla2tools`. The headline benchmark (Table 1) ran on a dedicated 16-vCPU x86_64 cloud host with 32 GB RAM, Ubuntu 24.04 LTS, no co-tenant workload. All harness sources, the patch, candidate-database snapshots, prompt templates, and the runbook are released as the verification artefact.

Workload. Three logical clients, one preferring each node of a three-node cluster, issue mixed reads (30%) and writes (70%) with per-client $p_{\text{issue}} = 0.3$ per tick, gated on a 200-tick per-operation timeout. Per-tick fault injection probabilities and the message delay/duplication parameters are as in §3.3.

Seeds. The DST sweep uses a deterministic set of seeds $\{1, \dots, N\}$ with $N = 8$ per candidate evaluation. For the calibration measurements in §3 we use $N \in \{50, 100, 200\}$ at 2000 ticks.

Benchmark. We invoke `go test -bench=BenchmarkProposal3Nodes -benchtime=1s -count=5` and report the minimum ns/op across the five runs; the minimum is more robust than mean or median against background system-load excursions.

LLM budget. The evolution loop is bounded by a per-run iteration count and a per-iteration wall-clock cap. The frontier coding model is accessed through an OpenAI-compatible chat endpoint; we report token spend and total wall-clock alongside results in §6.

6 Results

6.1 Calibration and Noise Floors

On the unmodified upstream library at the pinned commit, the invariant oracle of §3.3 reports 0 violations across 200 seeds at 2000 ticks each with the full fault catalogue enabled. With faults disabled, the linearizability check passes 29 of 30 seeds (97%); with faults enabled, the pass rate falls to 83 of 100 seeds. The 17% false-positive rate under faults is attributable to tick-level read-resolution timing (multiple entries applying in one Ready cycle with overlapping write/read intervals) and leader-change races during in-flight reads; the harness documents this floor and uses it as the threshold below which linearizability-check failures are not treated as candidate rejections in the loop. We treat the invariant oracle as the strict gate.

The benchmark exhibits a strongly bimodal ns/op distribution under repeated runs of an unmodified binary. A 100-sample calibration on the evaluation host (Appendix A.3) gives a fast-mode minimum of 4905 ns/op against a median of 8972 and a p90 of 10734, a spread driven by system load; the in-loop evaluator reports the minimum across five `-benchtime=1s` runs, a choice we revisit in §6.3. The in-loop score divides by a per-host baseline from a coarser earlier calibration (2400 ns/op arm64, 4688 ns/op x86); 4905 is a later 100-sample recalibration, and as a common divisor the baseline does not affect rankings.

6.2 Loop Output

We ran the loop for 30 iterations over ~28 minutes of wall-clock on a single commodity arm64 host (in-loop baseline 2400 ns/op). Of 30 proposed candidates, *all 30 cleared the rejection oracle*: every candidate compiled, passed 8 fault-injected DST seeds at 1500 ticks, and produced a non-violating client history. None were rejected at the linearizability layer (the gate operates at the calibrated noise floor of §3.4). The mean per-iteration score was 1.067; 25 of 30 candidates (83.3%) scored at or above the calibrated baseline, with 5 candidates regressing (min = 0.795). The best score observed during the loop was 1.1516 at iteration 10 (program identifier 1c3a355f; bench evaluator $n_{\text{op}} = 2084$ ns under min-of-five sampling).

The strict invariant gate fired zero times across all 30 candidates' fault-injection sweeps. This is not a claim of generator infallibility; it is a claim about the LLM's tendency, under our prompt scoping, to produce edits inside a code region where most plausible mutations preserve the function's contract.

Reproducibility. An independent second 30-iteration run on the x86 droplet (fresh sampling, identical configuration and gate) reproduced the gate behaviour exactly: all 30 proposed candidates again cleared compilation, the 8-seed fault-injected DST sweep, and the linearizability layer, with zero invariant violations. We use this replicate for the scoring analysis in §6.3 and the ablation in §6.4.

6.3 Best Accepted Variant

The best accepted variant (ranked by paired median) introduces three coordinated edits to `maybeSendAppend`:

1. **Throttled early-bail:** the function computes `throttled := pr.State == StateReplicate && pr.Inflights.Full()` once at entry; when `throttled && !sendIfEmpty` the function returns immediately, skipping the storage-backed `raftLog.term()` lookup and all subsequent log access.
2. **Skip payloadsSize on empty entries:** the common heartbeat-style empty-`MsgApp` path bypasses `payloadsSize([])` entirely.
3. **Local cleanup:** `pr.Next` and `r.raftLog.committed` are hoisted into stack-locals, and the now-dead post-loop `if err != nil` branch is removed.

We re-validated the variant against three independent oracles. The full upstream test suite passes on all six packages (`go test ./...`). A 50-seed harness sweep at 2000 ticks each, with the full fault catalogue, reports 0 invariant violations. Finally, a paired 10-run benchmark using `-benchtime=2s` on a dedicated commodity cloud host between back-to-back runs of `raft.go.baseline` and the

Table 1: Paired 10-run BenchmarkProposal3Nodes on a dedicated 16-core Intel host: ns/op (lower is better).

	min	p25	median	max
baseline	5683	8678	9405	11439
candidate	6127	6416	7020	9448
improvement	-7.8%	+26.1%	+25.4%	+17.4%

candidate yields the distribution in Table 1. The candidate is substantially faster across the body of the distribution: 25.4% *median improvement* (9405 $\rightarrow$ 7020 ns/op), 26.1% *at p25*, 17.4% *at the max*. The candidate is slightly slower at the very minimum (-7.8%), reflecting a small fixed overhead from the extra throttle check; it more than compensates by collapsing the distribution’s long tail.

Evaluator-side score under-ranked the headline candidate. The per-iteration evaluator scored this candidate at ≈ 0.998 (a marginal regression) because its scoring function takes the min of five back-to-back bench runs. The bench has a wide bimodal distribution on this host; min-of- N is dominated by lucky-fast runs, where the candidate’s small fixed overhead shows. The paired-distribution analysis in Table 1 reveals that the candidate’s optimisations affect the tail far more than the floor, exactly the behaviour one wants in a consensus library, where p99 latency, not best-case throughput, governs end-to-end commit time.

The independent replicate of §6.2 makes the effect quantitative. Re-scoring its 31 evaluated candidates against the same calibration baseline, min-of- N counts only 3 as improvements while the paired median counts 11; more starkly, the two rules invert the ranking. The loop’s min-best candidate (90f41d5d, min-speedup 1.06) is a median *regression* ($0.98\times$, ranked 18th of 31 by median), whereas the genuine median-best (5866c047, $1.60\times$ on the median by collapsing the bimodal tail) is ranked only 9th by min and would be discarded as a regression (min-speedup 0.92). The full re-ranking is in Appendix A.5. A more robust evaluator would therefore score on the paired median rather than the min; we treat this as a methodological lesson and discuss it in §7.

6.4 Per-Layer Ablation

Across both 30-iteration runs the strict gate never fired, showing the constrained prompt is safe but not that the four layers are *non-redundant*. To measure layer-uniqueness we hand-wrote five faulty variants of maybeSendAppend, each breaking a single named contract, plus an unmodified control, and pushed every variant through all four layers at $N = 12$ seeds. The full outcome is in Appendix A.4 (Table 4); it has three parts.

First, *every* broken variant is rejected before it can be benchmarked, and the catches span all four layers: a nonexistent-method call at compile; an omitted progress update by the DST sweep (a seed-1 liveness timeout); a forged commit index by the linearizability checker; and a paused-follower flood and a prevIndex off-by-one by the upstream suite (the flood also tripped linearizability at 2/12 seeds, at the 1/12 noise floor, so L3 is what catches it reliably). Second, the layers are genuinely independent: the safety-critical commit-index lie is flagged non-linearizable on 10/12 seeds (far above the control’s 1/12 noise floor), yet *passes* the invariant oracle untouched, precisely the redundancy Layer 3 exists to provide. Third, the harness does not subsume the upstream suite: a self-correcting prevIndex off-by-one passes both DST layers untouched and the paused-follower flood clears them up to the noise floor, yet both are caught reliably by upstream unit tests, which is why we re-run that suite on accepted candidates (§6.3).

6.5 Cost

The 30-iteration loop completed in approximately 28 minutes wall-clock, dominated by per-iteration evaluation cost (median ~ 57 s; ~ 15 s LLM call, ~ 3 s upstream compile, ~ 8 s harness compile, ~ 1 s DST sweep at $N = 8$ seeds and 1500 ticks, ~ 30 s benchmark with five runs). Each iteration sends the full host file (the ~ 85 K-byte raft.go with markers) plus the system prompt as input to the LLM and receives a SEARCH/REPLACE diff back. Token spend per iteration is on the order of $2 \cdot 10^4$ input tokens and $5 \cdot 10^2$ output tokens, yielding a total spend bounded by approximately ten

311 US dollars at publicly listed inference prices for the frontier model class used. The harness, not the
312 model, dominates wall-clock; halving the benchmark time would roughly halve the loop’s run time
313 without materially affecting the result.

314 7 Limitations and Threats to Validity

315 **Oracle coverage is a lower bound.** The invariant oracle covers log-prefix agreement and per-term
316 leader uniqueness; it does not cover lease-based read safety, snapshot-and-truncation correctness, or
317 membership-change conflict avoidance. The linearizability check uses a single-register state machine.
318 The fault catalogue covers partition, drop, reorder, and duplication; crash-restart and slow-disk are
319 scheduled extensions.

320 **Synthetic performance, harness-only acceptance.** All benchmarks are against an in-memory
321 simulated network and `MemoryStorage`; we make no claim about production traffic or physical
322 contention. Harness-acceptance is necessary but not sufficient for upstream merge; mainline review
323 and shadow deployment remain in scope.

324 **Evaluator scoring and generator noise.** $\min\text{-of-}N$ benchmark scoring during evolution is jitter-
325 sensitive and demonstrably misranked our best candidate; paired-median scoring (Table 1) is more
326 robust and should be the in-loop metric in future runs. The coding model is stochastic and closed-
327 source; different prompts, temperatures, or checkpoints will explore different regions.

328 8 Related Work

329 **Harness-first and verification-gated.** The methodological antecedents are the harness-first
330 `redis-rust` [23] and `Helix` [16] studies, and the Unicorn autonomous-optimization pipeline [5]
331 which gates LLM-evolved Rust behind a `Verus` [19] safety proof. Our work specialises the pattern
332 to a single function in a production Go library (no formal verification gate; an empirical DST oracle
333 in its place) and makes the harness the contribution rather than the implementation.

334 **Formal verification of distributed systems.** `IronFleet` [13] (Dafny) and `Verdi` [31] (Coq) gave
335 machine-checked proofs of distributed implementations at scale, and `Verus` [19] brings SMT-backed
336 verification to Rust. Go has no such analogue, so our Layer 1 is an existing TLA^+ trace validator
337 rather than a refinement proof.

338 **LLM-driven code synthesis and evolution.** The lineage runs from `Codex` [4] and `AlphaCode` [20]
339 to evolutionary search (`FunSearch` [26], `AlphaDev` [22], `AlphaEvolve` [3]). The ADRS pro-
340 gramme [6] applies the latter via an open-source framework [25] across ten benchmarks; we narrow
341 it to one safety-critical function. SWE-bench [15]-style agent benchmarks gate at the repository level,
342 not the function-and-trace level we use.

343 **Deterministic simulation and distributed-systems testing.** `FoundationDB`’s simulator [32,
344 33] and Jepsen [17] pioneered seeded simulation and linearizability checking, continued by
345 `Antithesis` [1], `TigerBeetle`’s `VOPR` [28], and `Madsim` [30], atop the property-based seeded-
346 random discipline [7, 21]. Our DST inherits `FDB`’s single-goroutine seeded-RNG style and the
347 register-history checker of `Knossos` [18]/`Porcupine` [2]; `etcd`’s robustness suite [12] targets the
348 integrated server, ours the algorithm library.

349 9 Conclusion

350 In a domain where safety properties are mechanically checkable, the harness is the load-bearing
351 artefact. Our pyramid gates an LLM-evolutionary loop over one function in `etcd-io/raft`; the
352 strict gate fires zero false positives across 200 seeds, and the headline accepted candidate delivers a
353 25.4% paired-median throughput improvement. The methodology generalises to any systems library
354 with a pluggable side-effect interface and a TLA^+ spec.

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459 A Technical Appendices and Supplementary Material

460 A.1 DST scheduler: full algorithm and bindings to upstream RawNode

461 Algorithm 1 gives the per-tick step in full; the remainder of this section explains how each line maps
462 onto the upstream RawNode/Storage interface.

Algorithm 1 One simulator tick (deterministic, single-threaded).

Require: cluster C , network N , seeded RNG ρ

```

1. Clocks.
1: for all healthy  $n \in C$  do
2:    $n.\text{TICK}(\rho)$  ▷ may fire election or heartbeat
3: end for
2. Let each node make progress.
4: for all  $n \in C$  do
5:    $\langle \text{entries}, \text{committed}, \text{outbox} \rangle \leftarrow n.\text{READY}()$ 
6:    $\text{PERSIST}(\text{entries}); \text{APPLY}(\text{committed}); \text{enqueue SORT}(\text{outbox}) \text{ into } N$ 
7:    $n.\text{ADVANCE}()$ 
8: end for
3. Deliver mail.
9: for all  $m \in N$  scheduled for this tick do
10:   $m.\text{dst.STEP}(m)$  ▷ may drop, reorder, or duplicate
11: end for
4. Answer clients.
12: for all pending read  $q$  with  $q.\text{idx} \leq \text{safeIndex}$  do
13:   complete  $q$ 
14: end for

```

463 **Phase 2 (Progress) bindings.** The destructure $\langle \text{entries}, \text{committed}, \text{outbox} \rangle$ corresponds to the
464 `rd.Entries`, `rd.CommittedEntries`, and `rd.Messages` fields of the `Ready` struct returned by
465 `RawNode.Ready()`. `PERSIST` writes to a per-node `MemoryStorage`; `APPLY` updates a single-register
466 state machine; `SORT` orders the outbox by $\langle \text{To}, \text{From}, \text{Type}, \text{Term}, \text{Index} \rangle$ before any message is en-
467 queued. The sort is mandatory: upstream constructs `r.msgs` by iterating a `map[uint64]*Progress`,
468 and Go intentionally randomises map iteration order, so any deterministic execution must impose a
469 canonical order at this seam. After Phase 2 the harness calls `n.Advance(rd)` to release the `Ready`
470 block, matching upstream’s contract.

471 **Phase 3 (Delivery) and randomness.** The network applies drop, reorder, and duplication faults
472 using the parameters of §3.3. All randomness (fault choices, partition/heal events, scheduled-delivery
473 delays, and upstream’s own election timeout) is sourced from a single seeded `*rand.Rand`; the latter
474 is exposed by the four-line patch of §3, which is the only modification to the upstream library.

475 A.2 Search-Space Identifiers

476 Table 2 maps each lever named in §4 to its upstream `etcd-io/raft` identifier at the pinned commit.

Table 2: The optimization levers in `maybeSendAppend` and their upstream symbols.

Lever (§4)	Upstream identifier
Throttle check	<code>pr.IsPaused()</code> , <code>pr.Inflights.Full()</code>
Replication state	<code>pr.State == StateReplicate</code>
Owed-entries slice	<code>r.raftLog.entries(pr.Next, r.maxMsgSize)</code>
Progress update	<code>pr.SentEntries(...)</code> , <code>pr.SentCommit(...)</code>
Snapshot fallback	<code>r.raftLog.term(pr.Next-1)</code> error path

A.3 Host Benchmark Calibration

Table 3 is the unmodified-baseline ns/op distribution on the evaluation host (16-vCPU x86_64, 100 samples: 20 cycles of `-count=5`). The gap between the minimum and the median is the bimodality that the in-loop min-of- N scoring exploits (§6.3).

Table 3: Baseline BenchmarkProposal3Nodes distribution (100 samples, ns/op; lower is better).

min	p10	p25	median	p75	p90	max
4905	5425	6502	8972	9329	10734	15153

A.4 Planted-Bug Ablation: Full Results

Table 4 expands §6.4. “inv” is the invariant DST sweep (seed of first violation, or – if none); “lin” is the Porcupine fail-count over 12 seeds; “tests” is the upstream root-package suite. The first layer to reject each variant is in the final column.

Table 4: Planted-bug ablation, full per-layer outcomes ($N = 12$ seeds). The control is the unmodified baseline.

Variant (broken contract)	compile	inv	lin	tests	First catch
control (none)	ok	–	1/12	pass	(passes)
nonexistent method call	FAIL	–	–	–	L0 compile
omit progress update	ok	seed 1 [†]	0/12*	FAIL	L1 invariants
advertise commit=lastIndex	ok	–	10/12	FAIL	L2 linearizability
ignore IsPaused	ok	–	2/12	FAIL	L3 upstream tests
prevIndex off-by-one	ok	–	0/12	FAIL	L3 upstream tests

[†] Liveness/throughput collapse surfacing as a sweep timeout, not a safety-invariant violation. * All 12 linearizability seeds also timed out (reported UNKNOWN), consistent with the throughput collapse. The control’s 1/12 linearizability failure is the noise floor of §3.4; only counts well above it (e.g. commit-lie’s 10/12) are real rejections.

A.5 Min- vs. Median-Ranking on the Replicate

Table 5 re-ranks the 31 evaluated candidates of the replicate run (§6.2) under the two scoring rules, scored against the calibration baselines (min 4905, median 8972 ns/op). The rules disagree both on the winner and on how many candidates count as improvements.

Table 5: Scoring-rule comparison on the replicate run’s 31 candidates. Speedup > 1 is faster than baseline.

Selected by	Candidate	min-speedup	median-speedup
min-of- N (in-loop)	90f41d5d	1.06	0.98
median-of- N	5866c047	0.92	1.60
improvements (≥ 1.0) under min-of- N			3/31
improvements (≥ 1.0) under median-of- N			11/31

A.6 Candidate-Generator System Prompt

The following is the verbatim `system_message` supplied to the candidate generator (the contracts of §4 are its *hard constraints*). It is reproduced unedited except that the specific model identity is withheld for anonymous review.

You are an expert in distributed systems and the etcd-io/raft Go library. Your task is to improve a single function, (`r *raft`) `maybeSendAppend`, which controls per-follower flow control, batching, and pipelining of `MsgApp`

```

500 messages on the Raft leader. The function appears between the markers
501 "# EVOLVE-BLOCK-START" and "# EVOLVE-BLOCK-END" in the file.
502
503 Hard constraints (any violation will cause your candidate to be rejected by
504 a layered verification harness - TLA+-derived invariants, deterministic
505 fault-injection simulation, linearizability checks):
506 * Preserve Raft safety: election safety, log matching, leader completeness,
507   state-machine safety.
508 * Keep the function signature exactly:
509   (r *raft) maybeSendAppend(to uint64, sendIfEmpty bool) bool
510 * Honour pr.IsPaused(); never send when paused.
511 * Always call pr.SentEntries(...) and pr.SentCommit(...) after a
512   successful r.send(MsgApp).
513 * Never send entries past pr.Next or beyond r.raftLog (use
514   r.raftLog.entries(pr.Next, r.maxMsgSize) or a slice of its result).
515 * Fall back to r.maybeSendSnapshot(to, pr) when r.raftLog.term(prevIndex)
516   returns an error.
517
518 Allowed directions for improvement:
519 * Adaptive batching: vary the per-message entry-count target using
520   pr.Match, pr.Next, r.raftLog.committed, or r.maxMsgSize.
521 * Smarter use of pr.Inflights (e.g. early returns when Inflights.Full()
522   but commit index advanced).
523 * Coalesce or suppress redundant empty MsgApp.
524 * Pre-filter on r.raftLog.committed when nothing has actually advanced
525   for this follower.
526 * Reduce allocations or pointer chasing inside the hot path.
527
528 You may NOT introduce new package-level globals, new goroutines, new locks,
529 or new I/O. The function must remain pure with respect to its inputs.
530
531 Output diffs in OpenEvolve's SEARCH/REPLACE format, scoped strictly to the
532 "# EVOLVE-BLOCK-START"/"# EVOLVE-BLOCK-END" region of the file.

```

Conference For AI Scientists 2026 - AI Involvement Checklist

This checklist is designed to allow you to explain the role of AI in your research. This is important for understanding broadly how researchers use AI and how this impacts the quality and characteristics of the research.

The checklist has two parts: **Research Stage Assessment** (items 1-4), where you rate the level of AI involvement and iteration effort at each stage of the research process, and **AI System Documentation** (items 5-6), where you provide free-text descriptions of the AI systems used and their observed limitations.

Research Stage Assessment

For items 1-4, give a score from the scale below that defines the role of AI in each part of the scientific process. The scores are as follows:

- **[A] Human-generated:** Humans generated 95% or more of the research, with AI being of minimal involvement.
- **[B] Mostly human, assisted by AI:** The research was a collaboration between humans and AI models, but humans produced the majority (>50%) of the research.
- **[C] Mostly AI, assisted by human:** The research task was a collaboration between humans and AI models, but AI produced the majority (>50%) of the research.
- **[D] AI-generated:** AI performed over 95% of the research. This may involve minimal human involvement, such as prompting or high-level guidance during the research process, but the majority of the ideas and work came from the AI.

For each research stage where AI was involved, i.e. where you selected **[B]**, **[C]**, or **[D]**, please also indicate the approximate level of iteration effort required using the provided iteration macros. Count substantive attempts, prompts, runs, agent trajectories, or course corrections; do not count minor wording edits.

- **[Low]** : approximately 1-10 substantive attempts.
- **[Medium]** : approximately 10-100 substantive attempts, with some failed attempts or course corrections.
- **[High]** : approximately 100+ substantive attempts, with substantial exploration, failures, or trial-and-error.
- **[NA]** : not applicable because AI involvement was **[A]**.
- **[Unclear]** : the authors cannot reasonably estimate.

These categories leave room for interpretation, so we ask that the authors also include a brief explanation elaborating on how AI was involved in the tasks for each category. Please keep your explanation to less than 150 words.

1. **Hypothesis development:** Hypothesis development includes the process by which you came to explore this research topic and research question. This can involve the background research performed by either researchers or by AI. This can also involve whether the idea was proposed by researchers or by AI.

Answer: **[A]**

Iteration Effort: **[NA]**

Explanation: The research question originated from the authors' day-to-day work with production systems, where AI-written code was bottlenecked at review because no verification layer existed, and using a second LLM to check LLM-generated code seemed self-defeating. This motivated a deterministic harness that could serve as a source of truth for iteratively improving existing systems. We chose `etcd-io/raft` because it already supplies one piece of that harness (a published TLA⁺ specification), and built outward from there. The idea and framing were human-generated; AI was not used to propose the hypothesis.

2. **Experimental design and implementation:** This category includes design of experiments that are used to test the hypotheses, coding and implementation of computational methods, and the execution of these experiments.

Answer: [C]

Iteration Effort: [High]

Explanation: The verification strategy was designed by the authors: after working with proof-based verification (e.g. Lean, TLA⁺) we judged formal specifications usable as a verification layer for AI-generated code, then added deterministic simulation testing (DST), which has a strong industry track record, and a Jepsen-style linearizability layer. Implementation was largely AI: the agentic coding assistant built the harness, the deterministic simulator, the evaluator, and the regression tooling under human direction and review, and a frontier LLM was the inner candidate generator inside the loop. Humans set the design, the search space, and the gate; the AI produced most of the code and all candidate edits.

3. **Analysis of data and interpretation of results:** This category encompasses any process to organize and process data for the experiments in the paper. It also includes interpretations of the results of the study.

Answer: [B]

Iteration Effort: [High]

Explanation: Data processing used AI-built tooling (the evaluator's distribution logging, the min- vs. median re-ranking script, and the planted-bug ablation harness), but interpretation was human-led and corrective. The agentic assistant initially summarized results with optimistic single-point statistics; the authors required paired benchmark distributions, explicit noise-floor characterization, and a min-of- N vs. median-of- N comparison, which materially changed the reported numbers (an apparent min-of-five speedup collapsed under paired-median measurement). The central interpretive conclusions, that the in-loop metric misranks candidates and that the harness rather than the model is load-bearing, were drawn by the authors from AI-generated tables.

4. **Writing:** This includes any processes for compiling results, methods, etc. into the final paper form. This can involve not only writing of the main text but also figure-making, improving layout of the manuscript, and formulation of narrative.

Answer: [C]

Iteration Effort: [High]

Explanation: The manuscript was drafted with the agentic coding assistant, which produced initial prose for most sections, the LaTeX tables, and figure/layout scaffolding. The authors set the narrative (methodology-first framing), the claims and their scope, and revised extensively: tightening wording, removing over-claims, and reconciling numbers against the experimental logs. Multiple rounds of human review corrected optimistic or imprecise statements before they reached the final text. AI produced the majority of the draft text; the authors own the framing, the final claims, and verification of every reported number.

AI System Documentation

For items 5-6, provide a free-text description.

5. **AI systems used:** Describe all AI systems used in this research without naming specific products or models. Use system-level descriptors only (e.g., "a frontier large language model", "a protein structure prediction system", "a custom multi-agent framework built on open-source models"). Include relevant details such as system type, scale, and capabilities. **Specific model and product names should only be included in the camera-ready version.**

Description: A single frontier large language model in agentic coding configuration, accessed through an OpenAI-compatible chat completion endpoint provided by a third-party inference platform. The same model identity was used both as (a) the agentic coding assistant that constructed the verification harness, the evaluator, the regression tooling, and this manuscript, and (b) the inner candidate generator inside the evolutionary optimization loop. An open-source evolutionary code-optimization framework derived from AlphaEvolve provided the MAP-Elites candidate database, parent sampler, and prompt-templating layer; we did not

633 modify the framework. The Porcupine linearizability checker is used as the consistency
634 oracle in the harness but is a deterministic algorithmic component, not an AI system.

635 6. **Observed AI Limitations:** What specific limitations and failure modes have you observed
636 when using AI as a partner or lead author?

637 Description: Three failure modes recurred. (1) *Method invention*: the inner candidate
638 generator occasionally proposed code referencing methods that do not exist on the relevant
639 structs (e.g. a hallucinated `IsHealthy` on the progress tracker); these were caught at the
640 compile layer of the harness rather than ending up in deployed code. (2) *Score-metric*
641 *over-fitting*: when the evaluator scored on a min-of-five benchmark, the generator produced
642 candidates whose “speedup” was largely the result of single-run jitter in the lower tail; the
643 apparent fifteen-percent improvement reduced to a paired-median four percent under more
644 rigorous measurement. (3) *Optimistic self-reporting*: when asked to summarise experimental
645 outputs, the agentic coding assistant initially reported the optimistic-end statistic; subsequent
646 prompts requiring paired distributions and explicit noise-floor characterisation produced
647 materially different numbers. We did not measure cross-model stability, and the underlying
648 inference service is stochastic; different runs of the same loop are not expected to produce
649 identical candidates.

Conference For AI Scientists 2026 - Reproducibility and Responsibility Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract and §1 enumerate four contributions (harness, calibration, evolutionary-search wrapper, empirical study) that map directly to §3, §6.2, §4, and §6.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: §7 discusses oracle coverage as a lower bound, the fault catalogue's omissions (crash-restart, slow disk), the synthetic-performance caveat, the harness-vs-deployment distinction, generator stochasticity, and benchmark noise floors.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [\[NA\]](#)

Justification: The paper does not include theoretical results. The Raft invariants we check come from prior published work [24] and the in-tree TLA⁺ specification; we do not extend or re-prove them.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [\[Yes\]](#)

Justification: §5 records the pinned upstream commit (etcd-io/raft @ 67d129e8, 2026-05-26), the Go and tooling versions (Go 1.26.3, OpenEvolve 0.2.27, Porcupine v1.1.0, OpenJDK 17), the hardware configuration (dedicated 16-vCPU x86_64 cloud host, 32 GB RAM, Ubuntu 24.04), the seeded fault parameters, the seed set, and the benchmark methodology (Go's standard `go test -bench=... -benchtime=1s -count=5` with min-reduction). The single upstream patch enabling deterministic seeded election timeouts is described in §3.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [\[Yes\]](#)

Justification: The harness source, evaluator, OpenEvolve configuration, prompt templates, and SQLite candidate database will be released as the verification artefact on acceptance, alongside the per-seed reproduction scripts referenced in §6. The release will be anonymised at submission and de-anonymised at camera-ready per CAISc policy.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [\[Yes\]](#)

Justification: §5 and §4 detail cluster size, tick budget, fault probabilities, scenario parameters, search space, generator temperature, max-tokens, and evaluator timeouts. The OpenEvolve config YAML is included in the artefact.

700 **7. Experiment statistical significance**

701 Question: Does the paper report error bars suitably and correctly defined or other appropriate
702 information about the statistical significance of the experiments?

703 Answer: [Yes]

704 Justification: §6.3 (Table 1) reports the min, median, and max of a paired ten-run benchmark
705 distribution rather than single-point speedups, and §7 explicitly characterises the bimodal
706 benchmark distribution and a $\sim 17\%$ linearizability false-positive noise floor below which ap-
707 parent improvements are not interpretable as real. We do not perform parametric hypothesis
708 testing because the underlying distribution violates normality.

709 **8. Experiments compute resources**

710 Question: For each experiment, does the paper provide sufficient information on the com-
711 puter resources (type of compute workers, memory, time of execution) needed to reproduce
712 the experiments?

713 Answer: [Yes]

714 Justification: §5 states the single-host commodity workstation context and the
715 embarrassingly-parallel-by-seed nature of the harness. §6.5 reports the per-iteration ~ 57 s
716 wall-clock decomposition, the ~ 28 -minute total run time, and the approximate ten-dollar
717 inference spend.

718 **9. Research Ethics**

719 Question: Does the research conducted in the paper conform with the [NeurIPS Code of](#)
720 [Ethics](#), except where CAISc’s AI authorship policies apply?

721 Answer: [Yes]

722 Justification: The research uses no human subjects, no personally identifiable data, and no
723 proprietary corpora. The AI Involvement Checklist documents the role of the language
724 model. The work targets a published open-source library; we do not introduce undisclosed
725 modifications and we propose no production deployment.

726 **10. Broader impacts**

727 Question: Does the paper discuss both potential positive societal impacts and negative
728 societal impacts of the work performed?

729 Answer: [Yes]

730 Justification: Positive: harness-gated agentic loops lower the cost of safety-preserving
731 optimisation in safety-critical infrastructure, and the verification-pyramid pattern generalises
732 to other distributed-systems libraries with published TLA⁺ specifications. Negative: the
733 same loop could be misused to produce subtly-incorrect code that nonetheless clears a
734 weak harness; we mitigate by emphasising in §7 that harness-acceptance is not deployment-
735 readiness and that the oracle’s coverage is a lower bound on bug-finding capability.